INNOPOLIS UNIVERSITY

Bachelor of Science in Computer Science

ViKi: Video-to-Kinematics for Object-Centric Retargeting of Human Manipulation Demonstrations to Robot Manipulators

Bachelor's Graduation Thesis

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Innopolis, Russia

2026

Abstract

ViKi is an offline research pipeline that transforms multi-view RGB-D recordings of tabletop manipulation into hand and object tracks and candidate robot trajectories. Its intended local task is a blue-cube transfer between two support cubes using two Azure Kinect DK cameras; the target embodiment is a simulated old-generation UR10 with a two-finger gripper. A geometric robot plan is not itself evidence of a physically successful manipulation.

The reproducible quantitative evaluation in this thesis uses a separate eight-view HO-Cap pick-and-place cohort: 28 sequences and 29,706 frames. Prepared hand tracks achieve 10.79 mm matched-joint-weighted MPJPE over 623,826 annotated joint slots. For 112 named object tracks, the macro-mean relative translation and rotation drift are 5.44 mm and 7.58 degrees after a first-frame reference alignment, conditional on published instance masks. Severe orientation outliers remain. The reference annotations are semi-automatic, and these scores do not validate the local Kinect rig or absolute object localization. Earlier local cube recordings and the artifacts derived from them are unavailable; their purported capture, retargeting, simulation and learning results are excluded. No physical-robot success, policy improvement or Real-to-Sim-to-Real transfer is claimed.

Keywords: imitation learning; motion retargeting; inverse kinematics; RGB-D calibration; object-centric representation; robot learning from human video.

Аннотация

ViKi — исследовательский офлайн-пайплайн преобразования многокамерных RGB-D записей настольной манипуляции в траектории кисти и объектов и кандидаты траекторий робота. Планируемая локальная задача — перенос синего кубика между двумя опорными кубиками с записью двумя Azure Kinect DK; целевой робот — моделируемый UR10 с двухпалым захватом. Геометрическая траектория сама по себе не доказывает успешную манипуляцию в физической симуляции.

Воспроизводимая численная оценка в работе выполнена на отдельном наборе HO-Cap: 28 последовательностей, 29 706 кадров и восемь камер. Взвешенная ошибка суставов кисти относительно полуавтоматической разметки составила 10,79 мм при 623 826 сопоставленных суставах. Для 112 поименованных треков объектов средняя по трекам ошибка относительного смещения составила 5,44 мм, поворота — 7,58 градуса после совмещения по первому кадру при использовании опубликованных масок. Сильные выбросы ориентации сохранены. Эти оценки не проверяют точность локального рига Kinect или абсолютную локализацию объектов. Прежние локальные записи с кубиками и производные артефакты недоступны для проверки; связанные с ними результаты записи, ретаргетинга, симуляции и обучения исключены. Работа не заявляет успеха на физическом роботе, улучшения политики или переноса Real-to-Sim-to-Real.

Ключевые слова: обучение с имитацией; ретаргетинг движения; обратная кинематика; калибровка RGB-D; объектно-центричное представление; обучение роботов по видео человека.

Acknowledgements

[Placeholder.] The author thanks **[supervisor]** for supervision and guidance throughout this work, and the Innopolis Robotics Society for maintaining a parallel development fork of the system, whose independent implementation attempts informed several of the architectural decisions reported here.

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Chapter 1

Introduction

1.1 Background

Learning manipulation from demonstrations is limited by the cost of acquiring robot-executable examples. Teleoperation records robot states and actions directly, but needs a robot and a control interface during collection. Passive RGB-D capture instead records a person's unconstrained motion with cameras. It is cheaper to collect, but it does not itself provide robot actions: hand motion must be reconstructed, related to objects, and transferred to a robot with different geometry. Even a plausible joint trajectory need not lift or place an object when contacts and dynamics are simulated.

This thesis studies that chain as a source of training data. The target experiment is deliberately narrow: a person transfers a blue 40 mm cube from a red support cube to a green target cube; a simulated old-generation UR10 with a Robotiq 2F-85 gripper attempts the same task. The goal is not to claim a general-purpose manipulation policy, nor to claim deployment on a physical robot that is not available for this study.

1.2 Problem Statement

The central question is whether passive demonstrations provide a measurable advantage when learning a robot policy in physics simulation. Three distinct gaps intervene. Perception must recover consistent hand and object motion from two imperfectly calibrated RGB-D views. Retargeting must turn a human motion into a reachable robot plan with a meaningful gripper command. Finally, the plan must survive physical execution: a kinematically valid approach may still miss a cube, slip during transfer, or fail on release.

Object-relative trajectories offer a way to transfer demonstrations to new cube placements, but they require a reliable object pose and must be checked again after transfer. Synthetic variations are therefore candidates for training, not valid demonstrations merely because a geometric transform can be computed. A second distinction is between learning with object coordinates and learning from

images. The former isolates the quality of transferred data and control; the latter also tests perception and the visual simulation gap.

1.3 Research Gap

Prior work covers parts of this problem. Teleoperation research develops retargeting objectives, while passive-video systems estimate human and object motion. MimicGen demonstrates that object-centric subtask transfer plus simulation filtering can expand a small source dataset [1]. This work does not claim those operators as novel. Its question is whether an accessible, camera-only capture pipeline can produce useful robot-learning examples for the specified cube-transfer task, and which stage limits the yield. The comparison must include a controller that uses no ViKi examples; otherwise success in simulation cannot be attributed to the capture pipeline.

1.4 Research Questions and Hypotheses

RQ. *Can passive two-view RGB-D demonstrations processed by ViKi improve learning of a simulated UR10 cube-transfer policy, compared with the same training task without ViKi-derived examples?*

RQ1. What fraction of recorded episodes survives hand reconstruction, object tracking, retargeting and physical replay, and where are the dominant losses?

RQ2. Does object-aware, physics-validated transfer to new cube placements improve held-out task success over equal-size world-frame or hand-only training sets?

RQ3. Under matched observations, control, training budget and evaluation placements, do ViKi-derived examples improve policy success or sample efficiency relative to learning without those examples?

RQ4. Can a visual policy using simulated camera frames and robot state retain task success without privileged object coordinates at test time?

These are testable hypotheses, not findings. H1 predicts a nonzero yield of physically successful transferred episodes. H2 predicts that object-aware examples improve success on unseen, reachable cube placements when example count is controlled. H3 predicts that prior ViKi examples reduce the number of simulation interactions needed to reach a given success rate. H4 predicts that a camera-based policy trained from a successful state-based teacher can perform the task when evaluated using only deployment-available observations. Failure at any gate is a result to report, not a reason to relabel a geometric plan as a physically successful demonstration.

1.5 Contributions

The implemented contribution is an offline pipeline from calibrated RGB-D recordings to hand trajectories, object tracks, and UR10 retarget plans. The experimental contribution sought here is a reproducible assessment of processing yield, simulated physical validity, object-relative variation, and the marginal value of ViKi-derived data for state and visual policies. Simulation replay, learning outcomes and real-world transfer are not claimed until measured. The proposed experiment uses no motion-capture ground truth or physical UR10; geometric consistency metrics must therefore not be presented as absolute pose accuracy.

1.6 Thesis Structure

Chapter 2 reviews capture, retargeting and demonstration-driven robot learning. Chapter 3 defines the implemented pipeline and the planned simulation study. Chapter 4 separates current evidence from experiments still to run. Chapter 5 treats uncertainty and validity, and Chapter 6 states the claims that the eventual results can and cannot support.

Chapter 2

Literature Review

2.1 Preamble

This chapter surveys the state of knowledge relevant to converting passively recorded human manipulation demonstrations into robot-executable datasets. It is organised not chronologically but by the architectural elements of the proposed system, so that for every design decision taken in Chapter 3 the alternatives considered and the reason for the choice are visible.

Section 2.2 examines how the human hand is represented, from full parametric meshes to the minimal wrist-pose-plus-gripper abstraction. Section 2.3 contrasts sequential and single-cost-functional retargeting formulations and reviews the solvers used to realise them. Section 2.4 compares object-centric and world-frame task representations. Section 2.5 covers multi-camera RGB-D calibration, depth fusion and cross-device synchronisation, with emphasis on reported quantitative accuracy. Section 2.6 reviews dataset formats, downstream policies and demonstration curation. Section 2.7 positions ViKi against the nearest published systems, and Section 2.8 states the resulting methodological choices.

Sources were drawn from arXiv (`cs.RO`, `cs.CV`, `cs.LG`), the proceedings of CoRL, RSS, ICRA, IROS, CVPR and NeurIPS, peer-reviewed sensor and calibration literature, and vendor documentation for the depth cameras and solvers employed. Where a source is a recent preprint whose peer-review status could not be confirmed, this is stated explicitly.

2.2 Representing the Human Hand

2.2.1 Full parametric reconstruction

A large body of work recovers the complete kinematics of the demonstrator. OKAMI [11] combines SMPL-H for the body with MANO for the hand and retargets the two separately. DexCanvas [24] builds a human–robot correspondence dataset by fitting MANO to data from twenty-two infrared

motion capture cameras and two RGB-D sensors. DexMV [12] likewise recovers a full MANO hand together with object pose from video before translating the demonstration into simulation.

Full reconstruction yields a rich representation—the pose of every finger joint—and is necessary when the target embodiment is a multi-fingered dexterous hand. It also introduces a heavy perception stack and an additional, poorly characterised error source: the mesh fit itself. For a parallel-jaw gripper, most of that representational richness is discarded downstream.

2.2.2 Minimal representation: wrist pose and gripper state

The alternative, adopted in this work, represents the hand as a wrist pose in $\mathrm{SE}(3)$ together with a scalar or binary gripper state. This is not a concession made for tractability; it is the representation used by the mainstream of current teleoperation systems. Open-TeleVision [3] maps the operator's wrist pose to the arm end-effector through closed-loop inverse kinematics and encodes a parallel gripper by the single thumb–index vector. Bunny-VisionPro [4] and ACE [5] adopt the same decomposition, ACE additionally demonstrating that it transfers across dexterous hands, parallel grippers and quadrupeds.

Methodological support for down-weighting the wrist comes from the ablation study of Xin et al. [8], who assemble a comprehensive retargeting objective—global hand pose, overall hand shape, relative fingertip positions and fingertip orientations—and remove terms one at a time. They find that precise tracking of the global wrist pose is frequently unnecessary and can degrade fingertip accuracy, and consequently replace the wrist-position objective with a thumb-tip objective while regularising wrist orientation only weakly. The implication for a wrist-plus-gripper representation is that the wrist target should enter the optimisation as a weighted soft task rather than a hard constraint.

2.2.3 Egocentric and monocular capture

Several systems reduce capture-hardware requirements further. EgoMimic [17] records bare-handed demonstrations through Project Aria glasses and co-trains on human and robot data. DexWild [18] collects in-the-wild bare-hand data through a low-cost portable device and likewise co-trains. A recent egocentric RGB-D system [23] generates trajectories offline using analytic damped-least-squares inverse kinematics and replays them on a low-cost arm. These works confirm a trend toward lighter capture rigs, but each accepts monocular or single-view geometry, and none combines it with the metric multi-view accuracy that the present work requires.

2.3 Retargeting Formulations

2.3.1 The single cost functional

The prevailing modern formulation poses retargeting as a constrained nonlinear least-squares or quadratic program over the robot configuration $\boldsymbol{q}_t$ at each timestep, minimising a weighted sum of a data-fidelity term and a temporal smoothness term subject to joint limits. AnyTeleop [2] gives the canonical instance: the objective sums, over keypoints, the squared residual between a scaled human keypoint vector and the corresponding robot forward-kinematics vector, plus a first-order penalty on configuration change between consecutive frames. The accompanying `dex-retargeting` library exposes a vector optimiser for real-time teleoperation, a position optimiser explicitly recommended for offline dataset processing, and a contact-aware DexPilot optimiser; it is built on Pinocchio and solved with sequential quadratic programming.

Two refinements are directly relevant here. First, per-term confidence weighting: in the formulation reported by Open-TeleVision [3], each keypoint residual is scaled by a distance-dependent switching weight that increases when a pinch is detected. The mechanism is precisely analogous to weighting by a per-frame perception confidence, and it establishes that weighting individual data terms by a scalar signal is accepted practice. Second, second-order smoothness: DexFlow [6] augments the standard objective with a dynamic smoothing weight and Hessian regularisation to enforce C^2 continuity, penalising curvature and not merely velocity. Robustification by a Huber loss on the data term, reported in recent dexterous teleoperation work [7], provides a mechanism for rejecting perception outliers inside the optimiser rather than smoothing them away beforehand.

2.3.2 The sequential pattern and its critique

Against this, the sequential pattern—filter the observed task-space trajectory, solve inverse kinematics per frame, filter the joint output— remains common in offline video-to-robot pipelines. Its deficiency is structural rather than empirical. Filtering is applied in task space, whereas the smoothness requirement is on the joint trajectory; because inverse kinematics is nonlinear, a smooth Cartesian input does not imply a smooth joint output, particularly near singularities. Pre-filtering also attenuates genuine fast motion indiscriminately with noise. In the single-functional formulation the smoothness penalty acts directly on $\boldsymbol{q}_t$, so joint-space smoothness is obtained by construction.

2.3.3 Offline pipelines and their retargeting operators

ORION [10] does not perform per-frame hand retargeting: it constructs an object-centric plan over sparse keyframes from an open-world object graph. OKAMI [11] warps a reference trajectory to a new object position and then solves inverse kinematics—the sequential pattern. HOWTransfer [20], the closest published pipeline in workflow terms, performs passive markerless capture and transfers

trajectories using flow-matching $\mathrm{SE}(3)$ grasp sampling with Laplacian trajectory editing, not a single weighted cost functional; its representation is hand-centric rather than object-centric. No offline pipeline located in this review adopts the AnyTeleop-style single functional.

2.3.4 Solvers

Pinocchio underlies the retargeting stack of AnyTeleop, Open-TeleVision and ACE. Pink [9] exposes weighted inverse kinematics directly as a quadratic program: each task contributes a quadratic residual with a weight matrix normalising task coordinates to common units, while joint limits, velocity limits and self-collision barriers enter as inequality constraints, optionally with control-barrier formulations. This structure admits the entire proposed cost functional natively—a frame task for the wrist target, posture and velocity regularisation tasks for smoothness, and limits as constraints— solved as one program per frame. The solver choice made in Chapter 3 therefore lies on the mainstream path rather than diverging from it.

2.4 Task Representation

Object-centric representations dominate recent work on cross-environment and cross-embodiment generalisation. ORION [10] represents a task as the three-dimensional object trajectory relative to initial and final object positions, arguing that this generalises across visual backgrounds, camera shifts, spatial layouts and novel object instances, and that full-body reconstruction of the demonstrator is unnecessary. SPOT [13] encodes each task as the $\mathrm{SE}(3)$ object-pose trajectory relative to the target explicitly in order to decouple embodiment-specific actions from sensory input, enabling learning from action-free human demonstrations, and conditions a diffusion policy on the object trajectory. Flow-based formulations [14] make an equivalent argument while relaxing the rigid-object assumption that pose-trajectory methods require.

OKAMI [11] occupies an intermediate position: the representation is demonstrator-centric, but object-aware warping adapts trajectories to new object positions—an implicit acknowledgement that pure world-frame replay fails when the object moves. Vi-RoIL [21] is object-centric by construction, expressing wrist trajectories and keyframe hand poses in the object frame, but recovers metric scale from a monocular view of a marker cube and evaluates only in simulation.

The trade-off is that object-centric representations require robust object pose or segmentation, and pose-trajectory variants assume rigidity, whereas world-frame trajectories are simpler but tied to the recording layout. Storing both, as proposed in Section 3.7, defers the choice to the downstream consumer at negligible storage cost.

2.5 Multi-Camera RGB-D Calibration, Fusion and Synchronisation

HO-Cap provides eight calibrated RealSense views and semi-automatic MANO hand–object annotations [31]. It is an external reference for the hand-joint stage, not an optical-motion-capture measurement of our two-Kinect rig. Camera selection on the evaluated sequences must be reported as pilot selection rather than held-out performance.

2.5.1 Fiducial choice

ChArUco boards combine the sub-pixel corner accuracy of a checkerboard with the robust identification and occlusion tolerance of ArUco markers. A 6×8 board provides up to thirty-five corners per view against four for a single AprilTag, and remains detectable under partial occlusion and at the image boundary where a plain checkerboard fails. Reported reprojection errors for ChArUco and ArUco calibration are typically below half a pixel. AprilTags are easier to detect at long range and oblique incidence and suffice for extrinsic-only estimation, but are ill-suited to intrinsic calibration owing to the small number of corners. Current practice is therefore to use ChArUco for intrinsics and either fiducial for extrinsics, preferring AprilTags for oblique overhead views.

2.5.2 Reported accuracy of multi-Azure-Kinect rigs

Romeo et al. [25] provide the cleanest direct comparison of calibration procedures for multi-Kinect rigs, reporting that three-dimensional procedures outperform two-dimensional chessboard approaches, with mean distances between corresponding point-cloud points of 21.4 mm at colour resolution and 9.9 mm at infrared resolution for a static scene, and 20.9 mm and 7.4 mm respectively for a dynamic scene; body-joint alignment error was 35.4 mm. Darwish et al. [26] calibrate five Azure Kinects against a ChArUco-coded cube, combining reprojection error with point-to-plane and patch-to-plane terms to compensate depth noise. Kim et al. [27] perform coarse registration from body tracking followed by feature-match refinement, retaining above ninety-five per cent inliers. Bu and Lee [28] use markerless coarse calibration with coloured ICP refinement and report—importantly for rig geometry—that ICP fails for near-opposing camera placements owing to insufficient point-cloud overlap.

Single-sensor specifications bound the achievable accuracy from below. The Azure Kinect DK is specified at a random error standard deviation of at most 17 mm and a typical systematic error below 11 mm plus 0.1 per cent of distance [33]; independent evaluation confirms these figures [29], and comparative work finds improved spatial and temporal accuracy relative to Kinect v2 beyond 2.5 m [30]. The Intel RealSense D435i is specified at depth accuracy below one per cent of distance [34]. The practical consequence is that centimetre-level, not millimetre-level, agreement should be

expected between calibrated views.

2.5.3 Depth fusion

Naive averaging of overlapping depth observations is inferior to confidence-weighted fusion. Weighting by distance and by the angle between the surface normal and the viewing direction—so that surfaces observed near perpendicular contribute more than obliquely observed ones—is established practice [37]. Learned confidence maps estimated from the depth signal itself are used to re-weight depth features before fusion [38], and surfel-based reconstruction attaches an explicit confidence value to each element and performs curvature-weighted registration [39]. Both sensor families used in this work expose per-pixel invalidation or confidence signals that can drive such weights directly.

2.5.4 Cross-device synchronisation

Azure Kinect devices support hardware synchronisation through daisy-chained or star-wired trigger cabling, with subordinate devices required to start before the master and depth exposures staggered to prevent laser interference; the vendor documentation states explicitly that equal timestamps between depth and colour streams, or between cameras, are not guaranteed [33]. RealSense devices provide an analogous inter-camera synchronisation mode, with the counter-intuitive property that timestamps appear aligned when hardware synchronisation is disabled and exhibit bounded drift when it is enabled, the drift itself serving as verification [34, 35].

The pitfall for a heterogeneous rig is that hardware clocks on different device families are not mutually comparable, since each counts its own cycles from its own epoch. Documented practice for such rigs is to stamp every arriving frame with a single host-side monotonic clock and to align streams by nearest timestamp in that common domain, accepting millisecond-scale precision [35]. Because Azure Kinect and RealSense triggers are mutually incompatible, this is the only available mechanism for aligning the two families, and it is the mechanism adopted in Section 3.4.

2.5.5 Hand–eye calibration

The OpenCV `calibrateHandEye` routine implements the Tsai–Lenz, Park–Martin, Horaud–Dornaika, Andreff and Daniilidis solvers. A comparative noise study [36] finds the Daniilidis, Chou and Park methods most robust to translation noise, with Daniilidis yielding the best translation estimates, while Tsai, Li and Lu degrade as rotation noise grows. Methods that decouple rotation from translation are faster but propagate rotation error into the translation estimate; Daniilidis solves both simultaneously through dual quaternions. Because the present system anchors cameras and robot to a shared ChArUco workspace frame, classical hand–eye estimation is largely bypassed; where an explicit camera-to-robot transform is required, Daniilidis is used.

2.6 Dataset Generation and Downstream Policies

LeRobot-compatible datasets together with ACT and Diffusion Policy are potential downstream consumers. The implemented default export is a NumPy trajectory bundle, not a replay-validated LeRobot dataset. OpenVLA-OFT [16] reports that parallel decoding, action chunking, a continuous action representation and $L1$ regression raise average success across four task suites from 76.5 to 97.1 per cent while increasing action-generation throughput twenty-six-fold, and is therefore a credible alternative consumer of the generated data.

Curation matters as much as generation. Demo-SCORE [15] trains a classifier on policy rollouts to filter low-quality demonstrations and reports an absolute success-rate improvement exceeding fifteen to thirty-five per cent over training on all demonstrations. Smoothness-based scoring of trajectories has been reported to yield comparable gains at a fraction of the data **[citation to be confirmed]**. This motivates treating dataset curation as an explicit pipeline stage rather than an afterthought, and—in the present design—eventually screening physical replay before export. This screening is not implemented in the current system.

2.7 Positioning of the Proposed System

Table 2.I compares ViKi against the nearest published systems along the six criteria that jointly define the proposed architecture: (C1) passive markerless capture without teleoperation hardware; (C2) multi-camera metrically calibrated RGB-D sensing; (C3) object-centric dense trajectory representation; (C4) retargeting as a single confidence-weighted cost functional; (C5) export of an imitation-learning-ready dataset; and (C6) physical replay used to validate the dataset before export.

TABLE 2.I
Comparison of the Proposed System Against the Nearest Published Work Along the Six Criteria Defined in Section 2.7

System	C1	C2	C3	C4	C5	C6
ViKi (proposed)	✓	✓	✓	✓	✓	✓
HOWTransfer [20]	✓	∼	—	—	∼	✓
Vi-RoIL [21]	∼	—	✓	—	∼	—
ORION [10]	✓	∼	✓	—	—	—
OKAMI [11]	✓	∼	∼	—	∼	—
EgoMimic [17]	✓	—	—	—	∼	—
DexWild [18]	∼	—	—	—	∼	—
Tool-as-Interface [19]	✓	∼	∼	—	∼	—
RoboPaint [22]	—	✓	∼	—	✓	✓
DexCanvas [24]	—	✓	—	—	∼	—

✓ = satisfied; ∼ = partially satisfied; — = not satisfied.

No located system satisfies all six criteria simultaneously. HOWTransfer [20] is closest in workflow, satisfying C1 and C6, but is hand-centric rather than object-centric, uses calibrated stereo RGB

rather than metric RGB-D, and does not employ a single cost functional. Vi-RoIL [21] is closest in representation, satisfying C3, but relies on a monocular view of a marker cube and is evaluated only in simulation. The intersection of the online joint-optimisation retargeting literature with the offline object-centric dataset-generation literature is therefore unoccupied, and it is that intersection which this work targets.

Several of the systems listed are recent preprints [20, 22, 23, 24] whose peer-review status could not be confirmed at the time of writing; their reported results are treated as provisional.

2.8 Conclusion of the Review

The review supports the following methodological choices, which Chapter 3 implements.

1. *Hand representation.* A wrist pose in $\mathrm{SE}(3)$ with a continuous gripper opening is adopted, following [3, 4, 5], and the wrist target enters as a weighted soft task following the ablation evidence of [8]. Parametric mesh reconstruction is rejected as unnecessary for a parallel-jaw target and as an additional error source.

2. *Retargeting.* A single weighted cost functional is adopted, following [2, 6], with confidence weighting after [3] and Huber robustification after [7]. The sequential smooth–IK–smooth pattern is rejected on the structural grounds set out in Section 2.3.

3. *Solver.* Pink on Pinocchio [9] is adopted, since it expresses the entire functional natively as a constrained quadratic program and is the solver family used by the reference teleoperation systems.

4. *Task representation.* Object-relative trajectories are adopted following [10, 13], and stored alongside workspace-anchored trajectories so that the choice is deferred downstream.

5. *Sensing.* ChArUco workspace anchoring is adopted; centimetre-level cross-camera agreement is assumed on the evidence of [25]; camera placement preserves substantial view overlap on the evidence of [28]; depth is fused with distance, incidence-angle and sensor confidence weights following [37, 38].

6. *Synchronisation.* The two Azure Kinects support hardware synchronisation, and the application uses a host-side monotonic clock for cross-device frame pairing [35]. Local timing accuracy cannot be re-evaluated without the source recordings.

7. *Curation.* Physical replay screening is a proposed validation stage, motivated by [15] and [20]; the current implementation does not perform it.

Chapter 3

Design and Methodology

3.1 System Overview

ViKi is an offline pipeline: demonstrations are recorded, processed and exported in distinct stages, with no requirement that any stage run at interactive rates. This is a deliberate design choice. Relaxing the real-time constraint that shapes teleoperation systems permits offline reconstruction and robot-plan inspection. Physical replay is a planned validation interface, not an implemented or evaluated stage.

The pipeline comprises six stages, shown in Fig. 3.1.

1. *Capture.* Synchronised RGB-D streams are recorded from a calibrated multi-camera rig while a demonstrator performs the task bare-handed.

2. *Perception.* Hand landmarks are detected per view, lifted to three dimensions using sensor depth, and fused across views with confidence weights to yield a wrist pose in $\mathrm{SE}(3)$ and a gripper state.

3. *Representation.* The wrist trajectory is expressed both relative to the manipulated object and relative to the ChArUco workspace anchor.

4. *Retargeting.* A robot joint trajectory is obtained by minimising a single weighted cost functional subject to kinematic and collision constraints.

5. *Replay validation (stub).* The interface exists, but no physical robot execution or replay-based acceptance is implemented.

6. *Export.* A NumPy trajectory bundle and JSON manifest can be written without implying physical validity; LeRobot export is optional and subject to additional requirements.

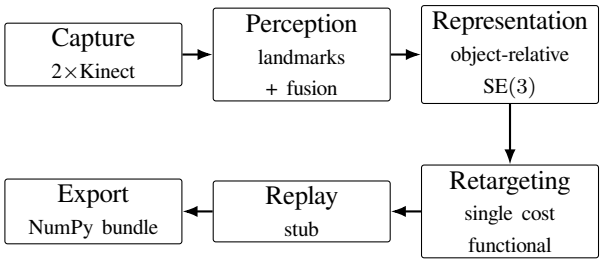


Fig. 3.1. Stages of the ViKi pipeline. Capture, perception and representation are embodiment-agnostic; retargeting is specific to the target manipulator. Replay is a stub, and export does not certify physical execution.

3.2 Hardware and Workspace

The local capture setup uses two Azure Kinect DK cameras observing a tabletop workspace. UR10 with a parallel-jaw gripper is the intended simulation target, not a physical manipulator mounted in this setup. Reachability of the unavailable local demonstrations cannot be quantified. The Azure Kinect time-of-flight depth characteristics are documented by the vendor [33] and studied independently [29]; neither source validates the current rig's calibration or fused trajectory accuracy.

The two Kinects are placed to preserve substantial overlap in their fields of view. This is a direct consequence of the finding of Bu and Lee [28] that iterative-closest-point refinement fails between near-opposing views for lack of point-cloud overlap; near-orthogonal placement retains both triangulation baseline and overlap.

3.3 Calibration

3.3.1 Intrinsics and extrinsics

Intrinsic parameters for every colour and depth stream are estimated from ChArUco board observations. Extrinsic calibration follows the workspace-anchor principle: rather than expressing cameras relative to one another, each camera is calibrated against a ChArUco board rigidly fixed to the workspace, defining a common world frame W. For camera k this yields

$$\boldsymbol{T}_{C_k}^{W} \in \mathrm{SE}(3), \tag{3.1}$$

the transform from the camera frame C_k to the workspace frame W. Anchoring to the workspace rather than to a reference camera has two consequences: adding or moving a camera does not invalidate the calibration of the others, and the robot base can be registered into the same frame without a separate camera-to-camera chain.

3.3.2 Robot registration

The manipulator base frame R is registered to W by observing a ChArUco target held in a known end-effector pose across a set of configurations, giving $\boldsymbol{T}_R^W$ directly. Where an explicit camera-to-robot transform is required instead, the Daniilidis dual-quaternion solver is used, following the robustness evidence of [36], with fifteen to twenty poses spanning a hemisphere.

3.4 Synchronisation

The two Azure Kinects are hardware-synchronised in master/subordinate configuration, with subordinate devices started before the master and depth exposures staggered to avoid mutual laser interference [33]. The RealSense cannot share that trigger; its synchronisation mode is incompatible at the electrical level.

Cross-family alignment therefore uses a single host-side monotonic clock. Every frame, from every device, is stamped on arrival with

$$\tau = \texttt{time.monotonic_ns()}, \tag{3.2}$$

and streams are aligned by nearest neighbour in τ. Two points deserve emphasis. First, per-device hardware timestamps are *not* used for cross-device arithmetic: distinct device families count their own cycles from their own epochs, and the vendor documentation states explicitly that equal timestamps across cameras are not guaranteed [33]. Second, a monotonic clock rather than wall-clock time is required, since wall-clock time may be adjusted discontinuously by time synchronisation daemons, which would corrupt interval computations mid-recording.

A future local evaluation must preserve paired device and host timestamps to measure alignment jitter and drift. No local synchronisation statistic is reported here; the HO-Cap benchmark cannot validate the Kinect rig.

3.5 Perception and Multi-View Reconstruction

3.5.1 Landmark detection and lifting

Hand landmarks are detected in each colour stream, yielding for view k at time t a set of image-plane landmarks with associated visibility scores $v_{i,t}^k \in [0, 1]$. Each landmark is lifted to three dimensions using the registered depth map rather than a learned depth estimate, exploiting the time-of-flight measurement directly, and transformed into the workspace frame by (3.1).

3.5.2 Multi-view triangulation

When a landmark is seen by two or more calibrated views, its position is not averaged from per-view depth lifts but estimated directly from the image observations, with depth entering as an additional soft residual rather than as the anchor. Averaging per-view lifts inherits the depth error of every view; triangulation is constrained by the camera geometry, which is known to calibration accuracy and does not degrade with range.

For landmark i at time t, let $\mathcal{K}_{i,t}$ be the views whose detector score clears a threshold and whose observation is finite. Every pair $(a, b) \subset \mathcal{K}_{i,t}$ yields a linear direct-linear-transform estimate, which is retained only if it lies in front of both cameras and if the two viewing rays subtend at least a minimum angle θ_{min}; a pair with a short effective baseline is ill-conditioned regardless of how confident the detector is. Each surviving hypothesis is scored against all views in $\mathcal{K}_{i,t}$ by the number of reprojection inliers, and the best is refined over its inlier set by

$$\hat{\boldsymbol{x}}_{i,t} \;=\; \arg\min_{\boldsymbol{x}} \sum_{k \in \mathcal{K}^{\text{in}}_{i,t}} \rho\Big(\big\| \pi_k(\boldsymbol{x}) - \boldsymbol{u}^k_{i,t} \big\|^2 \Big) \;+\; \lambda_d \sum_k \beta^k_{i,t}\, \rho_d\Big(\big(z_k(\boldsymbol{x}) - d^k_{i,t} - \delta_i \big)^2 \Big), \tag{3.3}$$

where π_k is the pinhole projection of view k, $\boldsymbol{u}^k_{i,t}$ the observed pixel, z_k the depth of $\boldsymbol{x}$ in that camera, $d^k_{i,t}$ the measured depth, and δ_i a per-landmark offset from the skin surface to the joint centre—depth sees the skin, the joint sits inside the finger. The depth weight $\beta^k_{i,t}$ is attenuated by the local depth spread and set to zero for invalid samples, so that a single bad silhouette pixel cannot move an otherwise well-constrained point. Observations are undistorted once, before the linear estimate, and every projection downstream is a plain pinhole map; distortion is never re-applied inside the residual.

The estimate carries a transparent geometric quality

$$q_{i,t} \;=\; \underbrace{\frac{|\mathcal{K}^{\text{in}}_{i,t}|}{|\mathcal{K}_{i,t}|}}_{\text{agreement}} \cdot \underbrace{\text{clip}\Big(1 - \frac{\bar{e}_{i,t}}{2e_{\text{in}}},\, 0,\, 1 \Big)}_{\text{reprojection}} \cdot \underbrace{\text{clip}\Big(\frac{\theta_{i,t}}{\theta_{\text{ref}}},\, 0,\, 1 \Big)}_{\text{baseline}}, \tag{3.4}$$

with $\bar{e}_{i,t}$ the mean reprojection error over inliers, $\theta_{i,t}$ the widest ray angle, and e_{in} the inlier radius. This is a transparent heuristic, not a calibrated variance, and it is used as such: it weights evidence, and is never interpreted as an inverse covariance.

A property of (3.4) deserves explicit statement, because it governs how much of a recording survives. The same e_{in} is the acceptance threshold, the scale of the robust loss ρ in (3.3), and the denominator of the reprojection factor. The factor is therefore designed to grade confidence out to $2e_{\text{in}}$, while the threshold admits nothing beyond e_{in}: half of the designed grading range is unreachable by construction. Widening e_{in} does not introduce a soft-weighting mechanism, it permits the existing one to operate over its own range. A consequence is that q is not comparable across settings of e_{in}: the same residual scores differently under a different denominator.

3.5.3 Single-view fallback

Where only one view observes a landmark, triangulation is undefined and the estimate falls back to the depth lift of that view. The historical multi-view path also combined per-view lifts by confidence-weighted averaging,

$$\hat{\boldsymbol{x}}_{i,t} \;=\; \frac{\sum_k w_{i,t}^k \, \boldsymbol{x}_{i,t}^k}{\sum_k w_{i,t}^k}, \qquad w_{i,t}^k \;=\; \underbrace{v_{i,t}^k}_{\text{detector}} \cdot \underbrace{c_{i,t}^k}_{\text{sensor}} \cdot \underbrace{\frac{1}{\left(d_{i,t}^k\right)^2}}_{\text{range}} \cdot \underbrace{\max\!\left(0, \cos\theta_{i,t}^k\right)}_{\text{incidence}}, \tag{3.5}$$

where $v_{i,t}^k$ is the detector score, $c_{i,t}^k$ the sensor depth validity at that pixel, $d_{i,t}^k$ the range, and $\theta_{i,t}^k$ the incidence angle. The range and incidence factors follow established practice in geometric depth fusion, where surfaces observed near perpendicular are weighted above obliquely observed ones because they are measured more accurately [37]; the detector and sensor factors follow confidence-map re-weighting [38]. The product form ensures that a zero in any factor—an undetected landmark, an invalid depth pixel, a grazing view— removes that view's contribution entirely rather than attenuating it.

This path is retained for reproducibility of earlier recordings and for single-camera operation. It is not the path used for the results reported here, because averaging per-view lifts propagates depth error into the fused point, whereas (3.3) constrains the point by geometry and admits depth only as evidence.

3.5.4 Wrist pose and gripper state

The wrist pose $\boldsymbol{T}_{H,t}^W \in \mathrm{SE}(3)$ is obtained by fitting an orthonormal frame to the fused wrist and knuckle landmarks. The gripper state is the thumb–index separation, mapped to a normalised opening,

$$g_t \;=\; \mathrm{clip}\!\left(\frac{\left\| \hat{\boldsymbol{x}}_{\text{thumb},t} - \hat{\boldsymbol{x}}_{\text{index},t} \right\| - \tau_{\text{closed}}}{\tau_{\text{open}} - \tau_{\text{closed}}},\; 0,\; 1 \right), \tag{3.6}$$

temporally filtered to suppress chatter. An earlier formulation thresholded the same distance into a binary open/closed indicator; it was replaced because a parallel-jaw gripper is a continuously commandable actuator, and quantising the one degree of freedom that carries the grasp discards precisely the approach and release phases that distinguish a successful demonstration from a failed one. This single scalar remains the entire hand representation beyond the wrist frame, following the convention established for parallel-jaw targets in [3, 5].

3.6 External Hand-Joint Reference Protocol

The main external cohort contains all 28 selected single-hand HO-Cap pick-and-place sequences (24 right-hand and four left-hand, 29,706 frames) from the pinned HO-Cap mirror revision recorded in the archived batch status. All eight RealSense RGB-D views are processed with MediaPipe observations, calibrated triangulation and the frozen `hocap-mediapipe-triangulated-v1` profile. The HoloLens stream is excluded. These sequences are separate from the local Kinect cube cohort and cannot validate its calibration. Published semi-automatic MANO 21-joint positions are transformed into the `hocap_tag_1` frame. Extract and Prepare do not use the reference hand labels. Predictions are paired by exact frame index and synthetic capture timestamp, without root alignment, Procrustes fit, temporal search or reference-guided filtering. MPJPE is the mean Euclidean error over finite matched joints; coverage is matched joints divided by valid reference joints. We report PCK@20/50 mm and the per-sequence 95th percentile alongside MPJPE. The aggregate MPJPE and PCK are weighted by matched joint counts; an aggregate P95 cannot be recovered from per-sequence quantiles. Incomplete Prepare stages are excluded.

For each of the four named objects in a sequence, a separate tracker receives *published instance masks* and depth from the eight views. It does not predict segmentation. Model confidence must reach 0.35. A single right-multiplied object-local SE(3) transform is fit at the first confident frame using the published object pose. Later matched frames are scored by translation distance and geodesic SO(3) angle in the same world frame; coverage is tracked frames divided by valid reference frames. Thus these are *relative tracking-drift* scores conditional on oracle masks and a reference-assisted first-frame gauge choice, not absolute 6-DoF localization. No symmetry-equivalent orientation metric or moving-only subset is applied. The 28-scene selection is a task-and-handedness subset, not an official held-out HO-Cap benchmark split. Earlier two-sequence, 120-frame two-versus- eight-view results remain a development pilot: its camera pair was selected on the same frames and cannot be treated as held-out evidence.

3.7 Object-Centric Representation

Let $\boldsymbol{T}_{O,t}^{W}$ denote the pose of the manipulated object in the workspace frame. Each demonstration is stored in two forms. The workspace-anchored form is $\boldsymbol{T}_{H,t}^{W}$ directly; the object-relative form is

$$\boldsymbol{T}_{H,t}^{O} \;=\; \left(\boldsymbol{T}_{O,t}^{W}\right)^{-1} \boldsymbol{T}_{H,t}^{W}. \tag{3.7}$$

The object-relative form is what transfers. If the same task is to be performed with the object at a new pose $\boldsymbol{T}_{O'}^{R}$ expressed in the robot frame, the desired end-effector pose at time t is

$$\boldsymbol{T}_{E,t}^{R,\text{des}} \;=\; \boldsymbol{T}_{O'}^{R}\, \boldsymbol{T}_{H,t}^{O}\, \boldsymbol{T}_{E}^{H}\,, \tag{3.8}$$

where $\boldsymbol{T}_E^H$ is a fixed offset relating the human wrist frame to the robot end-effector frame, determined once per embodiment. Trajectories are retained densely rather than reduced to keyframes, since the continuous velocity profile of the demonstration carries task-relevant information— approach speed, contact timing—that sparse keyframes discard.

Both representations are exported. Storing both costs little and defers to the downstream consumer a choice that the literature does not settle universally: object-relative trajectories generalise across object placement [10, 13] but depend on object pose estimation, whereas workspace-anchored trajectories are simpler and adequate within a fixed layout.

3.8 Retargeting as a Single Cost Functional

3.8.1 Formulation

Let $\boldsymbol{q}_t \in \mathbb{R}^n$ be the manipulator configuration at time t and $f(\boldsymbol{q}_t) \in \mathrm{SE}(3)$ the forward kinematics of the end-effector. The task-space residual with respect to the desired pose (3.8) is the logarithmic map of the pose error,

$$\boldsymbol{e}_t(\boldsymbol{q}_t) \;=\; \mathrm{Log}\Big(\Big(\boldsymbol{T}_{E,t}^{R,\mathrm{des}}\Big)^{-1} f(\boldsymbol{q}_t)\Big)^{\vee} \in \mathbb{R}^6. \tag{3.9}$$

Retargeting is then posed as the single constrained problem

$$\begin{aligned}\min_{\boldsymbol{q}_{1:T}} \;\; J(\boldsymbol{q}_{1:T}) \;=\; \sum_{t=1}^{T} \Big[& \underbrace{\omega_t\, \rho_\delta(\|\boldsymbol{e}_t(\boldsymbol{q}_t)\|_{\boldsymbol{\Sigma}})}_{\text{data fidelity}} \;+\; \underbrace{\lambda_v \,\|\boldsymbol{q}_t - \boldsymbol{q}_{t-1}\|^2}_{\text{velocity}} \\ & +\; \underbrace{\lambda_a \,\|\boldsymbol{q}_t - 2\boldsymbol{q}_{t-1} + \boldsymbol{q}_{t-2}\|^2}_{\text{acceleration}} \;+\; \underbrace{\lambda_r \,\|\boldsymbol{q}_t - \boldsymbol{q}_{\mathrm{ref}}\|^2}_{\text{posture}} \Big]\end{aligned} \tag{3.10}$$

subject to

$$\boldsymbol{q}_{\min} \;\le\; \boldsymbol{q}_t \;\le\; \boldsymbol{q}_{\max}, \tag{3.11}$$

$$\|\boldsymbol{q}_t - \boldsymbol{q}_{t-1}\| \;\le\; \dot{\boldsymbol{q}}_{\max}\, \Delta t, \tag{3.12}$$

$$h_j(\boldsymbol{q}_t) \;\ge\; 0, \quad j = 1, \dots, m, \tag{3.13}$$

where

ω_t	per-frame perception confidence weight, defined in (3.14);
ρ_δ	Huber loss with transition parameter δ;
$\mathbf{\Sigma}$	diagonal matrix normalising translational and rotational residual units;
$\lambda_v, \lambda_a, \lambda_r$	weights on velocity, acceleration and posture regularisation;
h_j	collision and self-collision barrier functions.

3.8.2 Confidence weighting

The per-frame weight aggregates the geometric qualities (3.4) of the landmarks that define the palm frame at time t,

$$\omega_t \;=\; \left(\frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} c_{i,t}\right)^{\alpha} \cdot \mathbb{1}[\text{frame valid}]\,, \qquad c_{i,t} = \begin{cases} q_{i,t}, & \text{landmark triangulated,} \\ 0, & \text{landmark interpolated,} \end{cases} \tag{3.14}$$

with $\mathcal{I}$ the four landmarks defining the palm frame—wrist and the index, middle and little metacarpophalangeal joints—and α controlling how sharply low-confidence frames are discounted.

Equation (3.14) as written admits a frame whose palm was only partially observed: a mean over four terms is non-zero whenever one of them is, so a palm with a single triangulated landmark still contributes at roughly a quarter weight, and the orientation it contributes was fitted partly through interpolated points. A stricter diagnostic variant zeroes the frame unless every landmark in $\mathcal{I}$ was triangulated,

$$\omega_t^{\text{strict}} \;=\; \omega_t \cdot \mathbb{1}[\,\forall i \in \mathcal{I} :\; i \text{ triangulated}\,]\,. \tag{3.15}$$

This variant is a proposed diagnostic, not a measured improvement in the available evidence. It trades observation support for stricter provenance; a retained local cohort is needed to quantify the effect on tracking and coverage. The construction is analogous to distance-dependent switching weights in online teleoperation retargeting [3]: poorly observed frames should exert less influence than well-observed frames.

3.8.3 Robustification

The Huber loss ρ_δ bounds the influence of gross outliers—a mis-detected hand, a depth dropout that survives the confidence weighting—at the level of the data term itself, rather than allowing a filtering stage to smear the outlier across neighbouring frames [7]. This is the mechanism that replaces pre-smoothing: outliers are down-weighted where they enter, not attenuated after they have already propagated.

3.8.4 Why a single functional

Equation (3.10) differs from the sequential alternative in one structurally important respect. The smoothness penalties act on $\boldsymbol{q}_t$, in the space where smoothness is required, and they are traded against data fidelity by the solver rather than applied blindly beforehand. In the sequential pipeline, task-space filtering cannot guarantee joint-space smoothness because the forward kinematics is nonlinear, and it cannot distinguish task-relevant fast motion from noise because it operates without reference to the data term. Here the trade-off is explicit: on high-confidence frames ω_t is large and the solution tracks the observation; on low-confidence frames the smoothness terms dominate and the solution interpolates.

Second-order smoothness, expressed by the acceleration term, follows the C^2-continuity argument of DexFlow [6]. A trajectory that is first-order smooth may still exhibit curvature discontinuities that appear as jerk on the physical manipulator and as label noise to a policy trained on the data.

3.8.5 Solution

The problem is solved with Pink on Pinocchio [9]. Each term of (3.10) maps onto a weighted task in the quadratic program—a frame task for the data term, posture and velocity regularisation tasks for the smoothness terms—while (3.11)–(3.13) enter as inequality constraints, with collision avoidance expressed through control barrier functions. Levenberg–Marquardt damping stabilises the solution near singularities.

Because the setting is offline, the problem is solved over the whole trajectory rather than frame by frame, so the acceleration term is well defined at every interior timestep and the solution is not constrained by the causality requirement that an online solver faces. The trajectory is initialised from the previous solution warm-started along t, and boundary conditions at $t = 1, 2$ are handled by replicating the initial configuration.

3.8.6 Interpolation

Where gaps remain after fusion—landmarks lost across all views—the trajectory is completed by cubic spline interpolation over $\mathrm{SE}(3)$ prior to optimisation. Dynamic filters that impose a motion model, such as Kalman or extended Kalman filters, are deliberately not applied: they introduce a prior about the dynamics that is subsequently re-estimated by the optimiser, and the interaction between the two priors is difficult to characterise. The optimiser already imposes the required regularity through λ_v and λ_a.

3.9 Replay Validation

Physical replay remains an interface stub: no robot execution, measured proprioception, contact outcome, or controller tracking residual is available. Kinematic plans must not be presented as replay-validated demonstrations. Simulator or physical validation would require recorded execution, contact checks, explicit success criteria, and rejection of failed episodes; these results are not part of the present evidence.

3.10 Dataset Export

The implemented default export writes a self-contained NumPy trajectory bundle and a JSON manifest. It does not imply physical replay or outcome screening. An optional LeRobot-format export has additional eligibility requirements and dependencies; neither format makes an unvalidated plan a physical demonstration.

3.11 Assumptions and Simplifications

The following assumptions delimit the scope of the work and are revisited in Section 5.4.

1. *Parallel-jaw target.* The hand is reduced to a wrist pose and a single continuous gripper opening (3.6). Multi-fingered dexterous targets would require the finger-level representation rejected in Section 2.2.

2. *The gripper body is not the hand.* The target is the midpoint between thumb and index tips, but a parallel-jaw gripper interposes a body of its own between the flange and that point, and the whole assembly must stay above the workspace plane. For tasks performed against the table surface the demonstrated pose can therefore be kinematically unreachable even when the arm's reach is ample. The size of this gap requires evaluation on retained source episodes.

3. *Rigid objects.* The object-relative representation (3.7) presupposes a rigid body with a well-defined pose. Deformable objects would require a flow-based representation [14].

4. *Single-arm, quasi-static tasks.* The task class is restricted to single-arm tabletop manipulation at moderate speed, where the quasi-static assumption underlying purely kinematic retargeting holds.

5. *Fixed workspace.* The ChArUco anchor is rigidly attached to the workspace and the camera rig is not moved within a recording session.

6. *Kinematic-only transfer.* Forces and contact dynamics are not estimated. Tasks whose success depends on force regulation are outside the scope.

Chapter 4

Implementation and Evaluation

4.1 Software and Experimental Setup

The Python implementation separates capture, calibration, perception, preparation, retargeting and export. FastAPI serves offline jobs and the browser inspection interface; MuJoCo tools reside in a separate image. Each stage writes a durable episode artifact. The `rec.npz` hand extraction, `cln.npz` prepared motion, `object_models.npz` scene model and `plan.h5` robot plan represent different stages and are not interchangeable evidence. Physical-robot replay is a stub and is not evaluated.

The local apparatus has two Azure Kinect DK cameras on independent USB controllers. The intended tabletop task is to transfer a blue cube from a red support to a green target, using a simulated old-generation UR10 with a Robotiq 2F-85 gripper. No physical UR10 is available for this study. The earlier local cube recordings and their derived artifacts are not available for verification, so no numerical result from that campaign is included in this manuscript. In particular, local capture-synchronisation, stage-yield, object-model, IK, contact-replay and learned-policy results are *not* claimed. A new, preserved local cohort would be required to evaluate that end-to-end chain.

4.2 External HO-Cap Hand and Object Evaluation

The full external experiment covers 28 single-hand HO-Cap pick-and-place sequences from nine subjects: 24 right-hand and four left-hand, with 29,706 frames from eight calibrated RealSense views. These are not the local Kinect cube recordings. The source revision and all 114 numerical JSON files are preserved in the paper's pre-retarget results snapshot with SHA-256 hashes. Capture and object protocols are documented in the accompanying dated logs. Both processing queues completed 28/28 scenes. This is a selected task cohort, not an evaluation on the official HO-Cap held-out split [31].

4.2.1 Hand-Joint Accuracy

Every one of the 623,826 valid reference joint slots has a finite prepared prediction. Matched-joint-weighted MPJPE is 10.79 mm and PCK@20/50 mm is 92.77/99.85%. The per-sequence MPJPE range is 9.26–12.85 mm; the per-sequence P95 range is 17.72–31.90 mm. No pooled P95 is inferred from these 28 quantiles. The four left-hand scenes have weighted MPJPE 11.60 mm, versus 10.65 mm for 24 right-hand scenes; handedness and subject are confounded here, so this is descriptive rather than a controlled difference. The fused-observed stage, before final temporal smoothing, matches 623,799/623,826 slots (99.996%) with weighted MPJPE 11.16 mm. The all-finite smoothed score includes 27 filled slots and should not be mistaken for direct observation coverage. Table 4.I summarizes the cohort; all 28 episode rows, including P95 and PCK, appear in Table B.I.

TABLE 4.I
Eight-view HO-Cap 21-joint accuracy against published MANO fits

Subset and stage	Scenes	Matched / GT	MPJPE (mm)	PCK@20 (%)
All, fused observed	28	623,799/623,826	11.16	91.68
All, smoothed all	28	623,826/623,826	10.79	92.77
Left, smoothed all	4	86,667/86,667	11.60	87.56
Right, smoothed all	24	537,159/537,159	10.65	93.61

4.2.2 Named Object Tracks and Outliers

Each scene contains four annotated objects, giving 112 object–sequence tracks across 15 object groups. The official HO-Cap object descriptors [32] are keyed by IDs in Table B.III; for example, `G18_2` is a projector remote, not an anonymous object. All four instances in the `G18` scenes are the green tea latte mix, projector remote, right shoe and left shoe. This identity mapping comes from the authors' `hocap_info.yaml`, rather than appearance-based naming.

The tracker covers 117,916 of 118,824 valid reference object frames (99.24%); 97/112 tracks have complete coverage. Across the 112 *track means*, the median and arithmetic mean relative translation errors are 4.74 and 5.44 mm; the corresponding rotation values are 3.47 and 7.58 degrees. Frame-count-weighted averages of the track means are 5.60 mm and 7.55 degrees. These are not pooled frame-level P95 values. The median of the 112 track P95s is 10.60 mm translation and 9.25 degrees rotation. First-frame alignment is fitted using the reference pose, so the numbers measure subsequent relative drift rather than absolute object-frame accuracy. The supplied instance masks remove segmentation errors from this test; static intervals remain mixed with manipulated intervals.

The worst rotation case is the projector remote in `subject_7/162803`: mean 112.60 degrees and P95 175.26 degrees despite 98.63% tracked coverage and 12.38 mm mean translation error. In another `G18` scene (`subject_1/165502`), the same object ID has 6.97-degree mean rotation error, and the other three objects in the failing scene have 1.10–1.34-degree means. Thus the anomaly is

TABLE 4.II
Selected HO-Cap object failures; mean and P95 are within-track errors

Sequence	Object and official descriptor	Cov. (%)	T mean	T P95	R mean	R P95
S7/162803	G18_2 projector remote	98.63	12.38	20.44	112.60	175.26
S5/112303	G15_2 dandelion tea	100.00	17.24	21.81	109.65	128.43
S2/202617	G09_3 toothpaste box	100.00	5.77	13.50	74.03	132.88
S7/193506	G02_4 sauce bottle	99.65	2.82	10.67	56.57	169.89
S2/163929	G19_1 electric screwdriver	48.80	5.75	15.41	10.55	50.95
S2/201942	G06_1 DVD	100.00	19.39	25.24	1.90	5.63

sequence/object-specific rather than a global frame rotation. A visual review showed plausible late object placement under an alternate orientation, but did not establish a unique root cause; the outlier remains in all aggregates. The dandelion tea (109.65 degrees) and toothpaste box (74.03 degrees in one scene, versus 13.82 and 6.89 degrees in two others) show that this is not a single isolated failure. A low translation error also does not certify orientation, as the sauce bottle demonstrates. Conversely, the DVD has the highest mean translation error, 19.39 mm, with only 1.90-degree mean rotation error. The electric screwdriver is a coverage failure at 48.80%, so its conditional error on visible frames must not stand in for whole-sequence accuracy. Tables B.II and B.III retain all 112 object scores and official identities for audit.

4.2.3 Historical Two-Sequence Camera-Pair Pilot

Before the full eight-view run, two subject-1 sequences were tested for 120 frames each with a pair selected after screening all 28 pairs on those same frames. Table 4.III is retained as a historical *in-sample* development comparison; its original local artifacts were lost. The full-sequence eight-view results above supersede it as the reproducible hand/object reference cohort, but do not turn the selected pair into a held-out baseline.

TABLE 4.III
Historical HO-Cap Hand-Joint Pilot: Selected Pair Versus Eight Views

Sequence	Views	Coverage	MPJPE (mm)	P95 (mm)	PCK@20 (%)
170650	selected pair	2,520/2,520	11.33	22.71	87.70
170650	all eight	2,520/2,520	11.34	22.17	87.50
170105	selected pair	2,520/2,520	11.67	22.80	89.80
170105	all eight	2,520/2,520	10.50	20.15	94.88

The selected pair also failed one colour–depth consistency check. No HO-Cap score establishes the accuracy of the local Kinect calibration or physical success of the proposed robot motion.

4.3 Downstream Evidence Boundary

The archived HO-Cap experiment evaluates hand reconstruction and first-frame-aligned object tracking on a separate RealSense capture system. It does not establish the accuracy of the local Kinect rig or produce a validated robot demonstration. Retarget plans, simulator contacts, task success and policy learning require their own retained inputs, protocols and numerical artifacts before they can be reported as results. This chapter therefore ends at the external perception evaluation rather than extrapolating an end-to-end success rate.

Chapter 5

Analysis and Discussion

5.1 What the Results Establish

The retained eight-view HO-Cap evaluation covers 28 selected pick-and-place sequences from nine subjects. Prepared 21-joint hand predictions match all 623,826 valid reference joint slots, with 10.79 mm matched-joint-weighted MPJPE against the dataset's published semi-automatic MANO fits. This is an external-system perception measurement, not evidence that the local Azure Kinect rig has the same accuracy. The object evaluation covers 112 named tracks, but it uses published instance masks and a reference-assisted first-frame SE(3) alignment. Its 5.44 mm translation and 7.58-degree rotation macro-means describe relative tracking drift conditional on those inputs, not absolute object localization or autonomous segmentation.

5.2 Error Sources and Reference Frames

The hand and object numbers cannot be added into an end-to-end geometric error budget. The hand score is an absolute comparison in the HO-Cap world frame; the object score removes a per-object initial gauge difference. Both refer to semi-automatic annotations rather than an independent metrology system. The object tracker also receives masks generated outside ViKi, and static intervals are included alongside manipulation. Thus a small average object drift need not imply a sufficiently accurate grasp pose during contact.

The outliers make this distinction operational. The projector remote `G18_2` has 112.60-degree mean rotation error in one sequence at 98.63% coverage, but only 6.97 degrees in another. The dandelion tea and toothpaste box have additional large orientation errors. The sauce bottle combines 2.82 mm mean translation error with 56.57-degree mean orientation error. These tracks remain in the aggregate rather than being removed as failures. Symmetry-aware orientation scoring was not used; the visual review of the remote did not uniquely diagnose the failure. Conversely, the electric screwdriver is tracked for only 48.80% of valid reference frames, so its conditional error does not

describe the missing half.

For the local two-Kinect system, camera geometry, RGB–depth alignment, exposure timing and table-height registration remain unquantified in this manuscript. The former local numerical summaries cannot be rechecked because their source recordings and derived artifacts are unavailable. Code paths and visually plausible clouds are not substitutes for a preserved evaluation protocol.

5.3 Statistical Validity and Attribution

The HO-Cap selection is a single-hand pick-and-place subset, not the official held-out benchmark split. Subjects and hands are unevenly represented: 24 right-hand and four left-hand sequences. Joint slots within the same frames and object frames within the same sequence are correlated, so their large counts do not imply that many independent trials. The two-sequence camera-pair pilot was selected on the same development frames and cannot be interpreted as held-out evidence. The main eight-view aggregate is descriptive for this selected cohort; no causal claim about the benefit of camera count, training data or robot embodiment follows from it.

No surviving local cube dataset supports the previous downstream figures. In particular, there is no verifiable local stage yield, simulated stacking rate, policy comparison or contact-safety result. The current manuscript therefore does not claim an end-to-end camera-to-simulation proof of concept or Real-to-Sim-to-Real transfer.

5.4 Scope and Implications

The supported contribution is a reproducible external perception assessment, with explicit object identities, coverage, outliers and metric limitations, alongside an implemented offline software pipeline. The intended cube-transfer experiment remains a research question rather than a completed result. A new preserved local cohort should include measurement-only controls: known geometry at table and carry heights, a marked asymmetric object for orientation, and disjoint human trials. Retained simulator rollouts and matched no-demonstration baselines would then be needed before reporting physical task success or learning benefit. No physical UR10 evaluation is part of the current evidence.

Chapter 6

Conclusion

6.1 Summary

ViKi implements an offline route from multi-view RGB-D capture to separate hand, object and candidate robot-plan artifacts. This thesis reports the retained, reproducible HO-Cap perception evaluation, not an end-to-end robot-learning result. The earlier local cube recordings and their downstream artifacts are unavailable, so their numerical summaries and derived simulation or policy claims are excluded.

6.2 Answers to the Research Questions

RQ1: pipeline yield. The external eight-view HO-Cap evaluation completed on 28 selected sequences and 29,706 frames. Its prepared hand tracks have 10.79 mm matched-joint-weighted MPJPE against the published semi-automatic reference, with 623,826/623,826 valid joints matched. For 112 named object tracks, macro-mean relative translation and rotation drift are 5.44 mm and 7.58 degrees after first-frame alignment and with published instance masks. Large orientation failures remain, including 112.60 degrees mean error on one projector-remote track. These data quantify external perception, not the local Kinect-to-robot pipeline yield or absolute object pose accuracy.

RQ2: object-aware transfer. No retained local demonstration and matched simulator evaluation supports a claim about physics-validated transfer to new cube placements. The question remains open.

RQ3: learning benefit. No reproducible comparison against a matched no-ViKi learner is available. The benefit of ViKi-derived examples or their sample efficiency is not established.

RQ4: visual policy. No evaluated image-conditioned robot policy is reported.

6.3 Claim Boundary and Next Validation

The current evidence supports a measurable HO-Cap hand-reconstruction result and a conditional, relative object-tracking result. It does not validate local camera calibration, complete cube-transfer demonstrations, collision-free grasping, physical simulator success, policy generalization or real-robot transfer. Re-recording a preserved local cohort with explicit calibration controls is necessary before testing those claims. Each subsequent retarget, physics and learning result must retain its input scenes, protocol and numerical outputs so it can be independently checked.

6.4 Concluding Remarks

The distinction between a reconstructed hand, a tracked object, a geometric robot plan and a successful manipulation is essential. Only the first two have quantitative external evidence in this manuscript; the latter stages remain work to evaluate.

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Appendix A

Notation

TABLE A.I
Symbols Used Throughout the Thesis

Symbol	Meaning
W	workspace frame, defined by the ChArUco anchor
C_k	frame of camera k
R	manipulator base frame
E	manipulator end-effector frame
H	human wrist frame
O	manipulated object frame
$\boldsymbol{T}_B^A$	rigid transform from frame B to frame A, element of SE(3)
$\boldsymbol{q}_t \in \mathbb{R}^n$	manipulator configuration at timestep t
$f(\boldsymbol{q}_t)$	forward kinematics of the end-effector
$\boldsymbol{e}_t$	task-space residual, (3.9)
ω_t	per-frame perception confidence weight, (3.14)
$q_{i,t}$	geometric quality of a triangulated landmark, (3.4)
$w_{i,t}^k$	per-view weight, single-view fallback, (3.5)
$v_{i,t}^k$	detector visibility score
$c_{i,t}^k$	sensor depth confidence
ρ_δ	Huber loss with transition parameter δ
$\lambda_v, \lambda_a, \lambda_r$	velocity, acceleration and posture weights
g_t	continuous gripper opening, (3.6)
τ	host-side monotonic timestamp, (3.2)
T	number of timesteps in an episode

Appendix B

Full HO-Cap Perception Results

Sequence labels use S<subject>/<HHMMSS> from the original HO-Cap identifier. The complete unrounded JSON and checksums are preserved in `paper/results/hocap/pre_retarget/`. Tables in this appendix can be regenerated with `python3 scripts/build_hocap_paper_tables.py`; the script refuses modified source hashes or an incomplete cohort. Object descriptors follow the official HO-Cap toolkit's `config/hocap_info.yaml` [32]. Hand P95 is per sequence; object P95 is per object–sequence track, never pooled across rows.

TABLE B.I
Eight-view HO-Cap hand results by sequence; PCK values are percentages.

Sequence	Frames	Matched	MPJPE	P95	PCK@20	PCK@50
S1/165502	676	14196	12.85	25.48	82.83	99.87
S1/165807	798	16758	12.10	24.18	84.35	99.79
S1/170105	831	17451	11.61	21.46	92.71	99.91
S1/170650	741	15561	12.55	23.78	86.36	100.00
S1/170959	874	18354	11.26	21.67	92.45	99.96
S2/201316	828	17388	10.77	21.23	93.48	99.95
S2/201942	1561	32781	11.86	23.79	86.87	99.95
S2/202617	1271	26691	11.06	22.30	91.43	99.99
S2/163929	1129	23709	10.48	20.43	94.11	99.99
S3/154531	1686	35406	9.26	18.09	96.95	99.98
S3/155008	1046	21966	9.45	17.72	97.89	100.00
S3/161209	1071	22491	9.71	18.75	96.10	99.92
S3/161937	1269	26649	10.33	19.48	95.74	99.79
S3/162327	878	18438	9.82	19.30	95.79	99.98
S3/162756	946	19866	9.56	17.79	98.23	100.00
S4/163223	939	19719	12.54	30.34	86.51	99.24
S4/164812	974	20454	10.57	20.61	94.22	99.99

Sequence	Frames	Matched	MPJPE	P95	PCK@20	PCK@50
S5/112303	702	14742	10.95	26.74	91.05	99.54
S6/110646	1292	27132	10.02	19.26	95.51	99.98
S6/111118	1272	26712	10.54	21.84	93.52	99.87
S6/112546	1311	27531	10.06	21.66	93.83	99.69
S7/193506	1136	23856	12.40	31.90	87.20	98.66
S7/162803	1092	22932	10.10	21.11	93.81	99.86
S8/180651	985	20685	11.88	24.84	89.59	99.88
S9/123403	992	20832	10.28	19.54	95.56	99.99
S9/123725	1134	23814	10.95	20.10	94.81	99.99
S9/124057	1136	23856	10.88	21.39	93.45	99.84
S9/125315	1136	23856	10.74	20.45	94.53	99.97

TABLE B.II

Per-object relative-pose tracking after one first-frame SE(3) alignment. T and R are translation in mm and geodesic rotation in degrees.

Sequence	Object	Cov. (%)	T mean	T P95	R mean	R P95
S1/165502	G18_1	100.00	2.68	3.75	1.32	1.97
S1/165502	G18_2	100.00	7.70	24.15	6.97	27.41
S1/165502	G18_3	100.00	1.49	6.97	1.32	7.71
S1/165502	G18_4	100.00	1.66	7.76	2.20	5.86
S1/165807	G15_1	100.00	2.41	10.53	1.58	3.85
S1/165807	G15_2	100.00	2.00	2.76	3.02	3.34
S1/165807	G15_3	100.00	2.17	10.08	1.24	7.77
S1/165807	G15_4	100.00	6.02	12.56	10.65	18.60
S1/170105	G10_1	100.00	0.67	3.56	1.55	10.44
S1/170105	G10_2	100.00	5.57	6.37	11.96	20.04
S1/170105	G10_3	100.00	10.13	23.41	3.54	9.66
S1/170105	G10_4	100.00	1.23	4.46	1.06	3.36
S1/170650	G04_1	100.00	1.41	4.74	1.64	3.84
S1/170650	G04_2	100.00	2.21	3.05	1.08	2.63
S1/170650	G04_3	100.00	8.89	20.10	6.02	14.70
S1/170650	G04_4	100.00	0.72	2.98	0.32	2.15
S1/170959	G19_1	100.00	2.41	8.12	4.74	15.42
S1/170959	G19_2	100.00	12.85	46.05	3.41	8.96
S1/170959	G19_3	100.00	2.95	10.80	5.50	19.73
S1/170959	G19_4	100.00	14.36	28.21	13.02	65.53

Sequence	Object	Cov. (%)	T mean	T P95	R mean	R P95
S2/201316	G05_1	100.00	4.73	12.06	3.30	9.27
S2/201316	G05_2	100.00	0.94	4.74	0.74	4.64
S2/201316	G05_3	100.00	9.40	28.11	7.81	17.53
S2/201316	G05_4	100.00	6.03	12.08	4.18	12.11
S2/201942	G06_1	100.00	19.39	25.24	1.90	5.63
S2/201942	G06_2	100.00	0.55	4.18	0.67	7.04
S2/201942	G06_3	100.00	6.42	10.68	7.26	17.51
S2/201942	G06_4	99.68	7.40	28.59	13.03	40.35
S2/202617	G09_1	100.00	11.27	24.44	3.91	18.56
S2/202617	G09_2	100.00	3.93	8.17	3.14	6.62
S2/202617	G09_3	100.00	5.77	13.50	74.03	132.88
S2/202617	G09_4	99.84	13.28	31.71	5.18	12.13
S2/163929	G19_1	48.80	5.75	15.41	10.55	50.95
S2/163929	G19_2	100.00	9.67	25.64	1.94	5.60
S2/163929	G19_3	100.00	2.13	7.22	6.38	8.89
S2/163929	G19_4	100.00	10.74	34.86	2.20	6.96
S3/154531	G09_1	100.00	5.93	13.05	2.77	9.75
S3/154531	G09_2	100.00	6.10	13.35	6.79	28.73
S3/154531	G09_3	100.00	9.03	14.95	13.82	142.95
S3/154531	G09_4	100.00	7.41	17.99	4.77	7.02
S3/155008	G10_1	100.00	1.59	4.16	1.93	7.28
S3/155008	G10_2	100.00	5.55	12.66	3.94	17.99
S3/155008	G10_3	100.00	5.67	7.58	1.70	6.88
S3/155008	G10_4	100.00	4.51	8.15	2.31	9.23
S3/161209	G16_1	100.00	1.91	4.31	5.78	6.44
S3/161209	G16_2	100.00	4.74	11.53	13.93	90.87
S3/161209	G16_3	100.00	2.86	7.87	1.92	10.59
S3/161209	G16_4	100.00	9.37	20.60	9.55	41.50
S3/161937	G19_1	100.00	3.11	16.36	7.25	51.05
S3/161937	G19_2	100.00	5.98	14.12	1.52	5.72
S3/161937	G19_3	100.00	3.50	7.80	7.81	18.02
S3/161937	G19_4	100.00	18.47	45.93	4.65	12.46
S3/162327	G20_1	100.00	0.99	2.36	0.93	2.31
S3/162327	G20_2	100.00	4.66	10.04	3.04	7.04
S3/162327	G20_3	100.00	1.93	9.06	0.62	2.82
S3/162327	G20_4	95.22	6.53	14.24	6.58	21.72
S3/162756	G21_1	100.00	2.51	4.55	2.18	3.26

Sequence	Object	Cov. (%)	T mean	T P95	R mean	R P95
S3/162756	G21_2	100.00	3.91	14.02	8.22	52.38
S3/162756	G21_3	100.00	10.18	22.65	2.89	6.76
S3/162756	G21_4	99.89	4.70	10.28	4.46	29.13
S4/163223	G16_1	100.00	5.17	23.66	5.12	25.23
S4/163223	G16_2	100.00	3.78	9.71	4.00	9.65
S4/163223	G16_3	100.00	3.58	5.05	10.68	15.81
S4/163223	G16_4	93.93	9.75	28.59	9.55	35.11
S4/164812	G21_1	100.00	1.72	3.03	1.23	3.54
S4/164812	G21_2	100.00	2.86	10.76	6.74	16.99
S4/164812	G21_3	100.00	15.06	32.43	3.60	10.91
S4/164812	G21_4	100.00	7.72	13.79	5.29	19.97
S5/112303	G15_1	100.00	1.12	5.40	1.91	13.57
S5/112303	G15_2	100.00	17.24	21.81	109.65	128.43
S5/112303	G15_3	100.00	5.28	8.30	8.28	12.34
S5/112303	G15_4	100.00	4.63	11.75	8.16	27.42
S6/110646	G05_1	100.00	2.76	6.17	1.83	5.31
S6/110646	G05_2	100.00	1.77	3.99	0.69	1.72
S6/110646	G05_3	91.33	5.27	21.45	4.69	28.75
S6/110646	G05_4	98.22	7.22	17.21	9.47	24.79
S6/111118	G06_1	99.37	10.02	18.20	2.08	7.79
S6/111118	G06_2	100.00	1.49	2.64	5.78	8.36
S6/111118	G06_3	100.00	1.33	3.50	2.61	6.06
S6/111118	G06_4	96.31	8.01	13.28	41.23	65.31
S6/112546	G22_1	99.31	6.76	15.41	8.75	82.09
S6/112546	G22_2	100.00	1.77	4.50	1.49	2.46
S6/112546	G22_3	100.00	0.60	2.39	0.95	2.31
S6/112546	G22_4	100.00	10.98	30.93	2.42	7.98
S7/193506	G02_1	100.00	7.51	8.67	2.30	5.40
S7/193506	G02_2	99.65	4.85	7.52	4.64	11.53
S7/193506	G02_3	100.00	1.10	5.58	1.12	6.37
S7/193506	G02_4	99.65	2.82	10.67	56.57	169.89
S7/162803	G18_1	100.00	5.49	7.02	1.33	4.87
S7/162803	G18_2	98.63	12.38	20.44	112.60	175.26
S7/162803	G18_3	100.00	0.98	6.86	1.10	6.94
S7/162803	G18_4	100.00	1.64	4.22	1.34	4.97
S8/180651	G06_1	100.00	15.36	37.92	2.05	6.07
S8/180651	G06_2	100.00	0.87	4.45	0.77	3.83

Sequence	Object	Cov. (%)	T mean	T P95	R mean	R P95
S8/180651	G06_3	100.00	3.04	7.80	2.23	5.96
S8/180651	G06_4	99.90	5.01	13.93	12.31	35.29
S9/123403	G07_1	100.00	5.00	9.78	4.49	18.84
S9/123403	G07_2	100.00	0.44	2.22	0.32	1.99
S9/123403	G07_3	100.00	1.05	3.60	2.45	8.26
S9/123403	G07_4	100.00	2.80	5.69	1.90	3.75
S9/123725	G09_1	100.00	11.53	26.72	3.78	14.00
S9/123725	G09_2	100.00	6.25	16.46	4.82	17.28
S9/123725	G09_3	100.00	5.51	13.62	6.89	36.18
S9/123725	G09_4	100.00	1.92	3.65	1.32	3.58
S9/124057	G11_1	100.00	1.95	6.34	2.31	4.28
S9/124057	G11_2	100.00	1.11	3.09	0.91	3.17
S9/124057	G11_3	100.00	0.81	7.19	0.40	2.49
S9/124057	G11_4	100.00	12.67	28.84	4.81	8.00
S9/125315	G21_1	100.00	0.83	4.39	0.64	4.58
S9/125315	G21_2	100.00	5.66	13.19	5.88	18.69
S9/125315	G21_3	100.00	5.53	12.10	1.77	4.42
S9/125315	G21_4	100.00	11.16	40.09	3.89	12.78

TABLE B.III

Official HO-Cap object identities for the evaluated groups; suffix order is _1 through _4.

Group	Official object descriptions in suffix order
G02	whole milk carton; cooked ham; chocolate drink powder; sauce bottle
G04	flapjack mix; chocolate fudge; herring fillets can; crackers box
G05	vegetable oil spread; body lotion; peanut chocolate box; sponge
G06	DVD; pumpkin creamer; yellow mustard; dish brush
G07	game controller; hot cocoa mix; dandruff shampoo; toy axe
G09	candy box; toy car; toothpaste box; chocolate syrup bottle
G10	soup mix; gel toothpaste; chocolate biscuit sticks; body wash
G11	coconut water; baby powder; baking soda; chocolate bar
G15	men's body wash; dandelion tea; cooking spray; joy controller
G16	toilet cleaner; laundry detergent; small coconut water; fabric softener sheets
G18	green tea latte mix; projector remote; right shoe; left shoe

Group	Official object descriptions in suffix order
G19	electric screwdriver; blue spatula; deodorant; ping pong paddle
G20	cappuccino mix; mustard bottle; toilet cleaner; dog toy bone
G21	moisturizing lotion; playing cards; pink spatula; blue brush
G22	chocolate bar; mayomust sauce; soup mix; gray spatula